

6 **(a)** Express $\frac{1}{(r-1)^2} - \frac{1}{(r+1)^2}$ as a single simplified fraction. **[2]**

(b) Hence determine the limit which $\sum_{r=2}^n \frac{r}{(r-1)^2(r+1)^2}$ converges to as $n \rightarrow \infty$. **[5]**